Pricing the Energy Asian/Basket Zero-Cost Collar

A Black-76 Treatment with Constant Volatility

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Overview

This note derives, step by step, two consistent ways to price the Asian/basket zero-cost collar (ZCC) described in the challenge, We model the two futures as log-normal and present two numerical methods for pricing it: an exact bivariate lognormal Monte Carlo, and a closed-form moment-matching formula. Notation follows Benth, Benth & Koekebakker (2008), *Stochastic Modelling of Electricity and Related Markets*, Chapter 9, adapted to a constant-volatility (Black-76) setting and to an average written on two futures contracts of different maturity.

1 Notation and Setup

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_u\}_{u\geq t}, \mathbb{Q})$, with $\mathbb{Q}$ the risk-neutral (pricing) measure and t the valuation date.

- $f(u,\tau)$: futures price at time u for delivery/maturity τ, as in Benth, Benth & Koekebakker (2008), Eq. (9.1).
- $\tau_1 < \tau_2$: the two contract maturities spanned by the averaging window (Section 2).
- $t_1 < \cdots < t_N \leq T$: fixing (averaging) dates; T: settlement date of the derivative.
- $\mathcal{K}_1 = \{1, \ldots, n_1\}$, $\mathcal{K}_2 = \{n_1+1, \ldots, N\}$: the fixing dates referencing τ_1 and τ_2 respectively, $n_2 = N - n_1$; weights $w_1 = n_1/N$, $w_2 = n_2/N = 1 - w_1$.
- $\theta_k = t_k - t$: year fraction from today to fixing k.
- $D(t,T)$: discount factor to T from the SOFR curve.
- $\sigma_1, \sigma_2 > 0$: constant volatilities of $f(\cdot,\tau_1)$ and $f(\cdot,\tau_2)$; $\rho \in (-1,1)$: constant correlation between the two driving Brownian motions.
- Φ: standard normal cdf.

Specialising the general multi-factor dynamics of Benth, Benth & Koekebakker (2008), Eq. (9.1), to one constant-volatility factor per contract (as in their treatment of two correlated commodities in Sect. 9.2.1, here recast in the geometric/lognormal setting of Sect. 9.1.1), the risk-neutral dynamics are

$$\frac{df(u,\tau_1)}{f(u,\tau_1)} = \sigma_1\, dW_1(u), \qquad \frac{df(u,\tau_2)}{f(u,\tau_2)} = \sigma_2\, dW_2(u), \qquad d\langle W_1, W_2\rangle_u = \rho\, du, \qquad u \geq t, \tag{1}$$

so that, for $u \geq t$,

$$f(u,\tau_i) = f(t,\tau_i)\exp\Big(-\tfrac{1}{2}\sigma_i^2(u-t) + \sigma_i\big(W_i(u) - W_i(t)\big)\Big), \qquad i = 1,2. \tag{2}$$

Lemma 1 (Cross-covariance of the two risk factors)**.** *For $t \le u \le v$,*

$$\mathrm{Cov}\big(W_1(u) - W_1(t),\, W_2(v) - W_2(t)\big) = \rho\,(u - t).$$

Proof. Write $W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\,\widetilde{W}_2$ with $\widetilde{W}_2 \perp W_1$ (Cholesky representation of two correlated Brownian motions). Then $\mathrm{Cov}(W_1(u) - W_1(t), W_2(v) - W_2(t)) = \rho\,\mathrm{Cov}(W_1(u) - W_1(t), W_1(v) - W_1(t)) + \sqrt{1-\rho^2}\,\mathrm{Cov}(W_1(u) - W_1(t), \widetilde{W}_2(v) - \widetilde{W}_2(t))$. The second term vanishes by independence; the first equals $\rho \min(u - t, v - t) = \rho(u - t)$. □

Remark. *Under the roll rule of Section 2, every fixing in $\mathcal{K}_1$ occurs strictly before every fixing in $\mathcal{K}_2$. Lemma 1 then shows that the covariance between a τ_1-fixing and a τ_2-fixing depends only on the variance accumulated* up to the τ_1-fixing*, not on how the τ_2-contract subsequently moves. Concretely, this means every cross-leg covariance term appearing below (Sections 5 and 6) can be written as a single sum indexed only by the τ_1-fixing dates, rather than as a full double sum over all pairs of τ_1- and τ_2-fixings — which is exactly what turns each cross-term into a closed-form expression.*

2 The Averaging (Roll) Rule

The pay-offs we are interested in are basket options that involve more than one futures. On each fixing date in the averaging period, the reference price is the settlement price of the *first-nearby* (front-month) futures contract, i.e. the listed contract with the nearest maturity still trading on that date. On the final fixing date(s) of the averaging window — those falling within the front contract's last trading days — market practice replaces the soon-to-expire, thinly traded first-nearby contract with the *second-nearby* contract as the reference price, since front-month liquidity typically dries up immediately before expiry. This produces exactly two maturities, τ_1 (first-nearby) and τ_2 (second-nearby), spanning the averaging window, with n_1 fixings referencing τ_1 and $n_2 = N - n_1$ referencing τ_2, as already encoded in $\mathcal{K}_1, \mathcal{K}_2$ above. The reference value of the derivative is the weighted average

$$A := w_1 A_1 + w_2 A_2, \qquad A_1 = \frac{1}{n_1} \sum_{k \in \mathcal{K}_1} f(t_k, \tau_1), \qquad A_2 = \frac{1}{n_2} \sum_{k \in \mathcal{K}_2} f(t_k, \tau_2), \tag{3}$$

where $w_i = n_i / N$ (for $i = 1, 2$) is simply the fraction of the N fixing dates in the averaging window on which the i-th futures contract supplies the reference price — i.e. the number of days the i-th contract fixes, divided by the total number of fixings — consistently with w_1, w_2 as defined above. Equation (3) is a basket written on two futures legs, in the spirit of the linear combination $S(t) = aS^A(t) + bS^B(t)$ used for spread/basket options in Benth, Benth & Koekebakker (2008), Eq. (9.25).

3 The Black-76 Formula for a Single Contract

Before pricing the basket it is useful to recall the classical result this note specialises and generalises, and which will also serve as the degenerate-case validation of both methods below.

Proposition 1 (Black-76, constant volatility)**.** *The price at $t \le T \le \tau$ of a call with exercise time T, strike K, written on $f(\cdot, \tau)$ with constant volatility σ, is*

$$C(t; T, K, \tau) = D(t, T)\big[f(t, \tau)\Phi(d_1) - K\Phi(d_2)\big],$$

$$d_1 = \frac{\ln\big(f(t,\tau)/K\big) + \frac{1}{2}\sigma^2(T - t)}{\sigma\sqrt{T - t}}, \qquad d_2 = d_1 - \sigma\sqrt{T - t}.$$

This is the $u \ge t$, single-maturity, constant-volatility case of Benth, Benth & Koekebakker (2008), Prop. 9.1 (there stated for a general, possibly time- and maturity-dependent, multi-factor volatility $\sigma_k(u, \tau)$).

4 Bivariate log-normal with market volatility

4.1 Basket forward and moneyness

Because futures are driftless martingales under $\mathbb{Q}$, today's forward value of the average is known model-free:

$$F_{avg} = w_1 f(t, \tau_1) + w_2 f(t, \tau_2). \tag{4}$$

Given a basket strike K, its moneyness is $m(K) = K/F_{avg}$.

4.2 Equivalent single-name strikes

Both pricing methods below (Sections 5 and 6) need a single volatility for each leg, σ_1 for $f(\cdot, \tau_1)$ and σ_2 for $f(\cdot, \tau_2)$, consistent with the strike K of the basket option actually being priced — exactly as pricing a plain vanilla option requires reading the volatility at the strike the option is struck at, not at an arbitrary point on the smile. The difficulty is that K is a strike on the *basket average* A, not on either individual futures price, so there is no strike to read directly off either individual smile. We resolve this by first translating the basket strike K into an equivalent strike on each leg, chosen to carry the same moneyness as K does for the basket, and then reading the volatility off each leg's calibrated smile at that translated strike. Concretely, we map the same log-moneyness onto each leg, i.e. we require $K_i(K)/f(t, \tau_i) = K/F_{avg}$ for $i = 1, 2$:

$$\boxed{K_1(K) = f(t, \tau_1)\,\frac{K}{F_{avg}}, \qquad K_2(K) = f(t, \tau_2)\,\frac{K}{F_{avg}}} \tag{5}$$

This is the natural mapping in a lognormal world: preserving $K/f(t, \tau_i)$ across legs is the same as preserving $\ln\big(K/f(t, \tau_i)\big)$, the number of log-moneyness units the basket strike sits away from each leg's own forward. It amounts to assuming that the two individual smiles move together in a mutually consistent, sticky-moneyness fashion — worth recording explicitly as a modelling assumption, since it is what allows a single basket strike to pin down a volatility on each leg separately. The volatilities used in both pricing methods below are then read off the two calibrated smiles at the mapped strikes,

$$\sigma_1 = \widehat{\sigma}_1\big(K_1(K)\big), \qquad \sigma_2 = \widehat{\sigma}_2\big(K_2(K)\big). \tag{6}$$

5 Bivariate Lognormal Monte Carlo

Under constant volatility, $\ln f(\cdot, \tau_i)$ is a Gaussian process: one can jump directly from t to each fixing date without any time discretisation. There is no discretisation bias; the only source of error is statistical, of order $1/\sqrt{M}$ in the number of paths M.

5.1 Covariance structure of the log-fixing vector

Define $\eta_k = \ln f(t_k, \tau_{i(k)}) - \ln f(t, \tau_{i(k)})$, where $i(k) \in \{1, 2\}$ denotes the leg of fixing k (i.e. $i(k) = 1$ if $k \in \mathcal{K}_1$, $i(k) = 2$ if $k \in \mathcal{K}_2$). By (2), the vector $\eta \in \mathbb{R}^N$ is Gaussian with

$$\mu_k = -\tfrac{1}{2}\sigma^2_{i(k)}\,\theta_k, \tag{7}$$

$$\Sigma_{ab} = \begin{cases} \sigma_1^2\,\theta_{a\wedge b} & a, b \in \mathcal{K}_1 \\ \sigma_2^2\,\theta_{a\wedge b} & a, b \in \mathcal{K}_2 \\ \rho\,\sigma_1\sigma_2\,\theta_a & a \in \mathcal{K}_1,\ b \in \mathcal{K}_2 \end{cases} \tag{8}$$

the mixed case following directly from Lemma 1, since $t_a < t_b$ whenever $a \in \mathcal{K}_1, b \in \mathcal{K}_2$.

5.2 Algorithm

1. Build $\mu \in \mathbb{R}^N$ and $\Sigma \in \mathbb{R}^{N\times N}$ as in (8) (once per recalibration date).
2. Compute the Cholesky factor $\Sigma = LL^\top$.
3. For $j = 1, \dots, M$: draw $Z^{(j)} \sim N(0, I_N)$ and set $\eta^{(j)} = \mu + LZ^{(j)}$.
4. Reconstruct prices: $f^{(j)}(t_k, \tau_1) = f(t, \tau_1)\, e^{\eta_k^{(j)}}$ for $k \in \mathcal{K}_1$; $f^{(j)}(t_k, \tau_2) = f(t, \tau_2)\, e^{\eta_k^{(j)}}$ for $k \in \mathcal{K}_2$.
5. Averages: $A_1^{(j)} = \frac{1}{n_1}\sum_{k\in\mathcal{K}_1} f^{(j)}(t_k, \tau_1)$, $A_2^{(j)} = \frac{1}{n_2}\sum_{k\in\mathcal{K}_2} f^{(j)}(t_k, \tau_2)$, $A^{(j)} = w_1 A_1^{(j)} + w_2 A_2^{(j)}$.
6. Payoffs: $\text{Call}^{(j)} = \max(A^{(j)} - K, 0)$, $\text{Put}^{(j)} = \max(K - A^{(j)}, 0)$.
7. Price $= D(t,T) \cdot \frac{1}{M}\sum_j \text{Payoff}^{(j)}$; standard error $= D(t,T) \cdot \widehat{\text{sd}}/\sqrt{M}$.

The whole simulation is a single matrix–vector product applied to all M paths at once; it should not be implemented as a per-path loop.

5.3 Variance reduction: exact geometric-average control variate

Let $a \in \mathbb{R}^N$ be the fixed weight vector with $a_k = w_1/n_1$ for $k \in \mathcal{K}_1$ and $a_k = w_2/n_2$ for $k \in \mathcal{K}_2$, and define the weighted geometric average G through

$$\ln G = w_1 \ln f(t, \tau_1) + w_2 \ln f(t, \tau_2) + a^\top \eta. \tag{9}$$

As a linear combination of jointly Gaussian components, $\ln G$ is *exactly* Gaussian, with

$$m_G = w_1 \ln f(t, \tau_1) + w_2 \ln f(t, \tau_2) + a^\top \mu, \qquad v_G^2 = a^\top \Sigma\, a, \tag{10}$$

so G is exactly lognormal with $F_{geo} := \mathbb{E}^{\mathbb{Q}}[G \mid \mathcal{F}_t] = e^{m_G + v_G^2/2}$. This yields an exact Black-76 price for the option on the geometric average, $C_{geo}(K) = D(t,T)\big[F_{geo}\Phi(d_1) - K\Phi(d_2)\big]$ with $d_1 = (\ln(F_{geo}/K) + v_G^2/2)/v_G$, $d_2 = d_1 - v_G$, used as a control variate:

$$\widehat{C} = \widehat{C}_{MC,\,arith} - \widehat{C}_{MC,\,geo} + C_{geo}, \tag{11}$$

computed on the same simulated paths for both averages (Kemna & Vorst, 1990, here extended to a two-leg basket).

Remark (Free model-free sanity check)**.** *By the AM–GM inequality, $A \geq G$ path by path, hence* $\text{Call}_{arith}(K) \geq \text{Call}_{geo}(K)$ *and* $\text{Put}_{arith}(K) \leq \text{Put}_{geo}(K)$ *for every model and every strike. A violation of either inequality signals an implementation bug.*

5.4 Required validation

Set $n_1 = 1$, $n_2 = 0$ (or vice versa), with the single fixing at T: the basket collapses to Proposition 1. Verify that the Monte Carlo price converges to it as M grows, with a $\pm 1.96\, SE$ band shrinking at rate $1/\sqrt{M}$, across several maturities.

6 Moment Matching

6.1 Mean of each leg

Since $f(\cdot, \tau_i)$ is a $\mathbb{Q}$-martingale, $\mathbb{E}^{\mathbb{Q}}[f(t_k, \tau_i) \mid \mathcal{F}_t] = f(t, \tau_i)$ for every k, so

$$\mathbb{E}^{\mathbb{Q}}[A_1 \mid \mathcal{F}_t] = f(t, \tau_1), \qquad \mathbb{E}^{\mathbb{Q}}[A_2 \mid \mathcal{F}_t] = f(t, \tau_2). \tag{12}$$

6.2 Covariance of two lognormal variables

For X, Y lognormal with $\ln X \sim N(m_x, v_x)$, $\ln Y \sim N(m_y, v_y)$ and $\mathrm{Cov}(\ln X, \ln Y) = c$,

$$\mathbb{E}[XY] = e^{\,m_x+m_y+\frac{1}{2}(v_x+v_y)+c}, \qquad \mathbb{E}[X]\,\mathbb{E}[Y] = e^{\,m_x+m_y+\frac{1}{2}(v_x+v_y)},$$
$$\Longrightarrow \qquad \mathrm{Cov}(X,Y) = \mathbb{E}[X]\,\mathbb{E}[Y]\big(e^{c}-1\big). \tag{13}$$

6.3 Variance of A_1 and A_2

Applying (13) to $f(t_a,\tau_1), f(t_b,\tau_1)$, with $c = \sigma_1^2\theta_{a\wedge b}$ and $\mathbb{E}[f(t_a,\tau_1)]\mathbb{E}[f(t_b,\tau_1)] = f(t,\tau_1)^2$,

$$\mathrm{Var}(A_1) = \frac{f(t,\tau_1)^2}{n_1^2}\sum_{a\in\mathcal{K}_1}\sum_{b\in\mathcal{K}_1}\Big[e^{\sigma_1^2\theta_{a\wedge b}}-1\Big]. \tag{14}$$

Grouping the double sum by $k = a\wedge b$ (there are $2(n_1-k)+1$ ordered pairs with minimum index k, $k=1,\dots,n_1$) gives the closed form

$$\boxed{\mathrm{Var}(A_1) = \frac{f(t,\tau_1)^2}{n_1^2}\sum_{k=1}^{n_1}\big[2(n_1-k)+1\big]\Big(e^{\sigma_1^2\theta_k}-1\Big)} \tag{15}$$

and, by the same argument on the second leg,

$$\mathrm{Var}(A_2) = \frac{f(t,\tau_2)^2}{n_2^2}\sum_{j=1}^{n_2}\big[2(n_2-j)+1\big]\Big(e^{\sigma_2^2\theta_{n_1+j}}-1\Big). \tag{16}$$

6.4 Covariance between A_1 and A_2

$$\mathrm{Cov}(A_1,A_2) = \frac{1}{n_1 n_2}\sum_{a\in\mathcal{K}_1}\sum_{b\in\mathcal{K}_2}\mathrm{Cov}\big(f(t_a,\tau_1), f(t_b,\tau_2)\big). \tag{17}$$

By Lemma 1, $c_{ab} = \rho\sigma_1\sigma_2\,\theta_{a\wedge b} = \rho\sigma_1\sigma_2\,\theta_a$, which does *not* depend on b. The inner sum over b therefore collapses to a factor n_2:

$$\boxed{\mathrm{Cov}(A_1,A_2) = \frac{f(t,\tau_1)\,f(t,\tau_2)}{n_1}\sum_{k=1}^{n_1}\Big(e^{\rho\sigma_1\sigma_2\theta_k}-1\Big)} \tag{18}$$

Remark. *Equation (18) is a direct consequence of the roll rule of Section 2: because every τ_1-fixing precedes every τ_2-fixing, the cross-leg covariance "freezes" at the end of the $\mathcal{K}_1$ window and is independent of how the τ_2-contract subsequently evolves.*

6.5 Moments of the basket and lognormal fit

$$M_1 := \mathbb{E}^{\mathbb{Q}}[A \mid \mathcal{F}_t] = w_1 f(t,\tau_1) + w_2 f(t,\tau_2) = F_{avg}, \tag{19}$$

$$\mathrm{Var}(A) = w_1^2\,\mathrm{Var}(A_1) + w_2^2\,\mathrm{Var}(A_2) + 2\,w_1 w_2\,\mathrm{Cov}(A_1,A_2). \tag{20}$$

Matching the first two moments of A to a lognormal variable Y with $\mathbb{E}[Y] = e^{m+v^2/2} = M_1$ and $\mathbb{E}[Y^2] = e^{2m+2v^2} = M_1^2 + \mathrm{Var}(A)$ gives

$$v^2 = \ln\!\left(1 + \frac{\mathrm{Var}(A)}{M_1^2}\right). \tag{21}$$

6.6 Closed-form price

$$C(t;T,K) = D(t,T)\big[M_1\,\Phi(d_1) - K\,\Phi(d_2)\big], \tag{22}$$

$$P(t;T,K) = D(t,T)\big[K\,\Phi(-d_2) - M_1\,\Phi(-d_1)\big], \tag{23}$$

$$d_1 = \frac{\ln(M_1/K) + v^2/2}{v}, \qquad d_2 = d_1 - v. \tag{24}$$

6.7 Consistency with the benchmark

With a single fixing at T on one leg only ($n_1 = 1$, $n_2 = 0$, $\theta_1 = T - t$),

$$\mathrm{Var}(A_1) = f(t,\tau_1)^2\big(e^{\sigma_1^2(T-t)} - 1\big), \qquad M_1 = f(t,\tau_1), \qquad v^2 = \ln\big(1 + \mathrm{Var}(A_1)/M_1^2\big) = \sigma_1^2(T-t), \tag{25}$$

i.e. $v = \sigma_1\sqrt{T-t}$: the formulas above collapse exactly to Proposition 1.

7 Solving for the Zero-Cost Put Strike

Let K_{call} be the (given) call strike of the collar.

1. Compute $K_1(K_{call}), K_2(K_{call})$ from (5), read σ_1, σ_2 off the smiles, and price the call once: $C^* = C(t;T,K_{call})$ from Section 6 (or Section 5).
2. Define $h(K) = P(t;T,K) - C^*$, where for *every* trial K:
   - recompute $K_1(K) = f(t,\tau_1)K/F_{avg}$, $K_2(K) = f(t,\tau_2)K/F_{avg}$;
   - re-read $\sigma_1(K_1(K))$, $\sigma_2(K_2(K))$ from the smiles;
   - recompute $\mathrm{Var}(A_1), \mathrm{Var}(A_2), \mathrm{Cov}(A_1,A_2)$ and v^2;
   - price $P(t;T,K)$ as in Section 6.
3. Solve $h(K_{put}) = 0$ by bisection (or Newton's method, using $\partial P/\partial K \approx D(t,T)\Phi(-d_2)$ with σ_i frozen for the derivative step), with initial bracket $[0, F_{avg}]$.
4. In practice h is increasing (the direct effect of K dominates the indirect effect through the smile), but this should be checked numerically rather than assumed — as noted for the analogous put-call parity argument in Benth, Benth & Koekebakker (2008), Sect. 9.2.2.

Alternative Modelling Approaches

The constant-volatility, two-factor lognormal setting developed above is a deliberately tractable starting point. Both pricing methods extend to the following richer, but computationally heavier, models; we mention them here only for reference, without pursuing the details.

1. **Jump-extended HJM dynamics.** Replace the pure-diffusion dynamics (1) with the jump-diffusion HJM framework for forward and swap curves of Benth, Benth & Koekebakker (2008), Ch. 6 and Sect. 9.1.2, calibrated to match observed smiles. Under this richer dynamics the arithmetic basket (such as the considered case) generally loses closed-form tractability and pricing falls back on Monte Carlo, while the geometric average typically remains the piece still amenable to semi-closed-form pricing via Fourier/characteristic-function methods, along the lines of Benth, Benth & Koekebakker (2008).

2. **Spot-driven models.** Model the underlying spot price $S(t)$ directly (e.g. Heston, 1993, NIG...), recovering each futures price through the usual cost-of-carry relation $f(t, \tau_i) = S(t)\,\Lambda(t, \tau_i)$ for a deterministic function Λ. The spot model is calibrated jointly to the vanilla smiles on both legs, and the basket is then priced by simulating the single calibrated spot process forward and reconstructing $f(\cdot, \tau_1), f(\cdot, \tau_2)$ along each path. You can consider the cost of carry as a risk factor to hedge.

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